Class 09: Exploratory analysis of Halloween candy

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Background

In this mini-project, you will explore FiveThirtyEight's Halloween Candy dataset. FiveThirtyEight, sometimes rendered as just 538, is an American website that focuses mostly on opinion poll analysis, politics, economics, and sports blogging.

So what is the top ranked snack-sized Halloween candy? What made some candies more desirable than others? Was it price? Maybe it was just sugar content? Were they chocolate? Did they contain peanuts or almonds? How about crisped rice or other biscuit-esque component, like a Kit Kat or malted milk ball? Was it fruit flavored? Was it made of hard candy, like a lollipop or a strawberry bon bon? Was there nougat?

Your task is to explore their candy dataset to find out answers to these types of questions - but most of all your job is to have fun, learn by doing hands on data analysis, and hopefully make this type of analysis less frightening for the future! Let's get started.

Importing Candy Data

First things first, let's get the data from the FiveThirtyEight GitHub repo. You can either read from the URL directly or download this candy-data.csv file and place it in your project directory. Either way we need to load it up with read.csv() and inspect the data to see exactly what we're dealing with.

```
# Save the file name as an object
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranl

# Read the CSV file into R and use the first column as row names
candy <- read.csv(candy_file, row.names = 1)

# Show the rows of the candy data
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

2.1 What is in the data set

The dataset includes all sorts of information about different kinds of candy. For example, is a candy chocolaty? Does it have nougat? How does its cost compare to other candies? How many people prefer one candy over another?

According to 538 the columns in the dataset include:

- chocolate: Does it contain chocolate?
- fruity: Is it fruit flavored?
- caramel: Is there caramel in the candy?
- peanutyalmondy: Does it contain peanuts, peanut butter or almonds?
- nougat: Does it contain nougat?
- crispedricewafer: Does it contain crisped rice, wafers, or a cookie component?
- hard: Is it a hard candy?
- bar: Is it a candy bar?
- pluribus: Is it one of many candies in a bag or box?
- sugarpercent: The percentile of sugar it falls under within the data set.
- pricepercent: The unit price percentile compared to the rest of the set.
- winpercent: The overall win percentage according to 269,000 matchups (more on this in a moment).

We will take a whirlwind tour of this dataset and in the process answer the questions highlighted in red throughout this page that aim to guide your exploration process. We will then wrap up by trying Principal Component Analysis (PCA) on this dataset to get yet more experience with this important multivariate method. It will yield a kind of "Map of Halloween Candy Space". How cool is that! Let's explore…

> **Q1.** How many different candy types are in this dataset?

There are 85 different candy types in this data set.

> **Q2.** How many fruity candy types are in the dataset?

There are 38 fruity candy types in the dataset.

```
# Count how many different candy types are in the dataset
# Each row represents one candy type
nrow(candy)
```

```
[1] 85
```

```
# Count how many fruity candy types are in the dataset
# The fruity column uses 1 for fruity and 0 for not fruity
sum(candy$fruity)
```

```
[1] 38
```

2.2 What is your favorite candy?

One of the most interesting variables in the dataset is winpercent. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

We can find the winpercent value for Twix by using its name to access the corresponding row of the dataset. This is because the dataset has each candy name as rownames (recall that we set this when we imported the original CSV file). For example the code for Twix is:

```r
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

An alternate way to do this is with the help of the dplyr package that we have seen previously:

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

> **Q3.** What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy other than Twix is "3 Musketeers." Its winpercent value is 67.60294.

```r
# I chose KitKat as my favorite candy other than Twix
candy["3 Musketeers", ]$winpercent
```

```
[1] 67.60294
```

Q4. What is the winpercent value for “Kit Kat”?

The winpercent value for “Kit Kat” is 76.7686.

```r
# Q4: Find the winpercent value for Kit Kat
library(dplyr)

candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
        winpercent
Kit Kat    76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The winpercent value for “Tootsie Roll Snack Bars” is 49.6535.

```r
# Find the winpercent for Tootsie Roll Snack Bar
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side-note: the skimr::skim() function

There is a useful skim() function in the skimr package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```r
# Install the skimr package
#install.packages("skimr")

library(skimr)

skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> **Q6.** Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the winpercent column looks like it is on a different scale than the other columns. Most columns are on a 0 to 1 scale, but winpercent ranges from about 22 to 88.

> **Q7.** What do you think a zero and one represent for the candy$chocolate column?

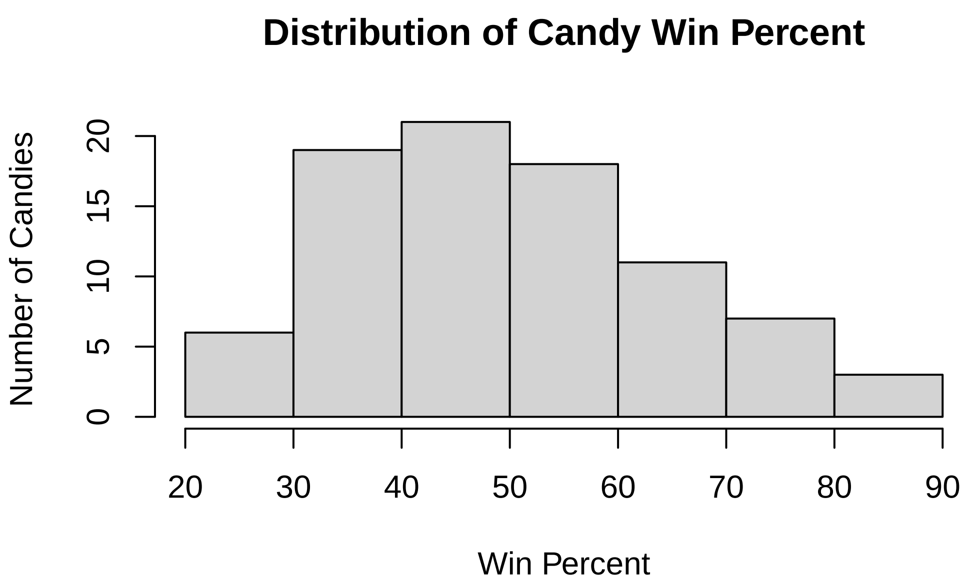
In the candy$chocolate column, 0 means that the candy is not chocolate. A value of 1 means the candy is chocolate.

Explanatory Analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function hist(). Alternatively, you can use ggplot() with geom_hist(). Either works well in this case and (as always) its your choice.

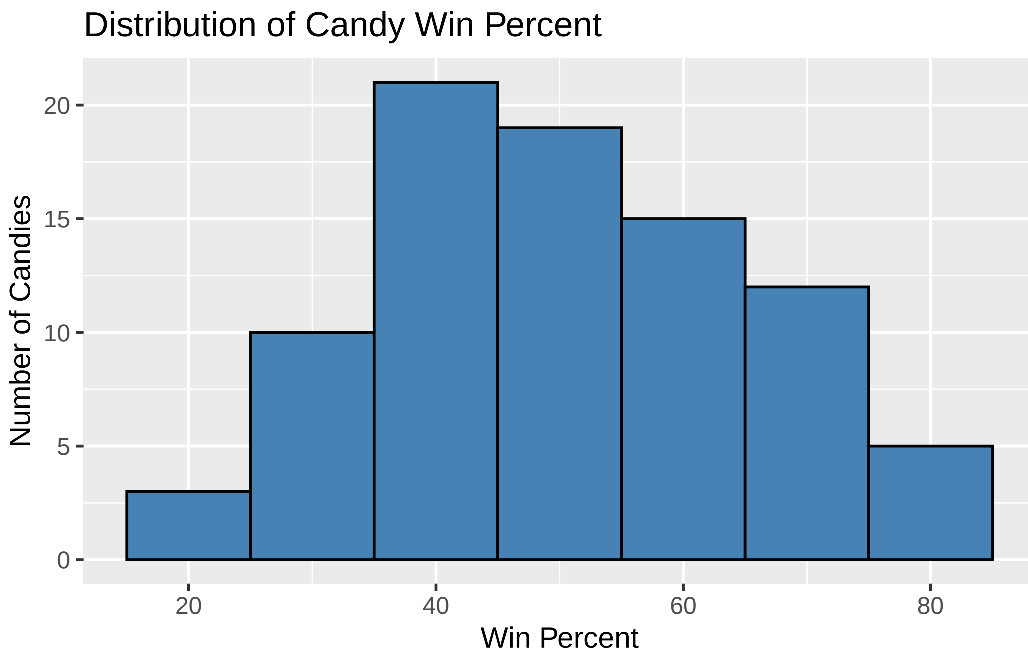
> **Q8.** Plot a histogram of winpercent values using both base R and ggplot2.

```r
# Q8. Plot a histogram of winpercent values using base R

hist(candy$winpercent,
     main = "Distribution of Candy Win Percent",
     xlab = "Win Percent",
     ylab = "Number of Candies")
```




```r
# Q8: Plot a histogram of winpercent values using ggplot2

library(ggplot2)

ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Candy Win Percent",
       x = "Win Percent",
```

```
       y = "Number of Candies",
       )
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not symmetrical, as it is slightly right skewed.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution is below 50% as most candies are clustered in the lower win percent range since the histogram is right skewed. From skim(), the median winpercent value is about 47.83%.

```
# Find the center of the winpercent distribution
median(candy$winpercent)
```

```
[1] 47.82975
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average, chocolate candy is higher ranked than fruity candy, as chocolate has a winpercent of 60.92153 and fruity candy has a winpercent of 44.11974.

```r
# Get winpercent values for chocolate candies
chocolate <- candy$winpercent[as.logical(candy$chocolate)]

# Get winpercent values for fruity candies
fruity <- candy$winpercent[as.logical(candy$fruity)]

# Calculate the average winpercent for chocolate candies
mean(chocolate)
```

```
[1] 60.92153
```

```r
# Calculate the average winpercent for fruity candies
mean(fruity)
```

```
[1] 44.11974
```

> **Q12.** Is this difference statistically significant?

Yes, the difference is statistically significant because the p-value is 2.871e-08, which is smaller than the 0.05 threshold.

```r
# Q12: Test if the difference between chocolate and fruity candy is statistically significant

# Use a t-test to compare the two groups
t.test(chocolate, fruity)
```

```

	Welch Two Sample t-test

data:  chocolate and fruity
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Overall Candy Rankings

Let's use the base R order() function together with head() to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the dplyr package you can use the arrange() function together with head() to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

The five least liked candy types in this set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```r
# Q13: Find the five least liked candy types

# Sort the candy dataset by winpercent from lowest to highest
head(candy[order(candy$winpercent), ], n = 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

Q14. What are the top 5 all time favorite candy types out of this set?

The top 5 all time favorite candy types out of this set are Reese;s Peanut Butter Cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

```
# Q14: Find the top 5 favorite candy types

# We sort the candy dataset by winpercent from highest to lowest
head(candy[order(candy$winpercent, decreasing = TRUE), ], n = 5)
```

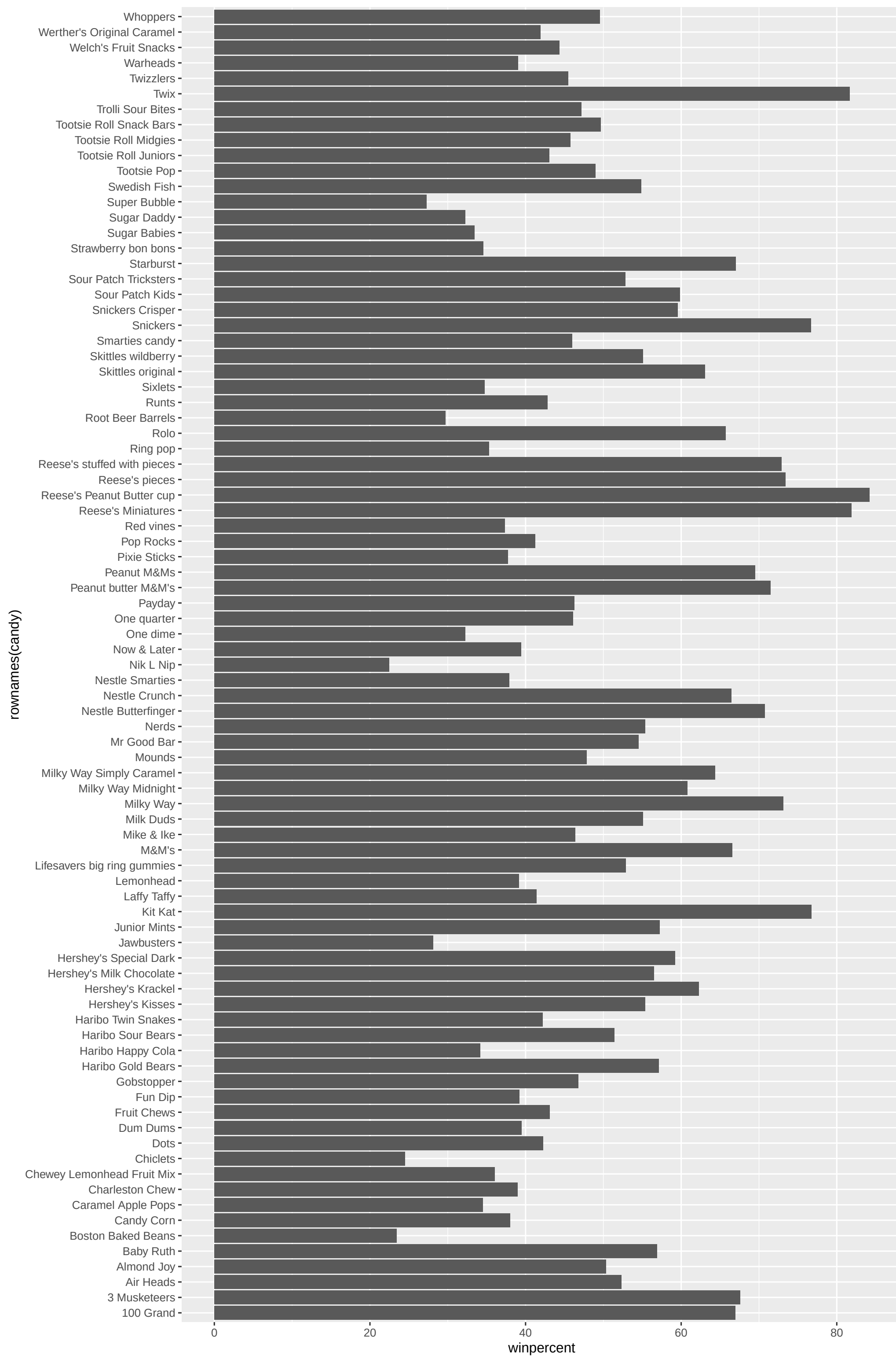
```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

To examine more of the dataset in this vain we can make a barplot to visualize the overall rankings. As always we will use an iterative approach to building a useful visualization by getting a rough starting plot and then refining and adding useful details in a stepwise process.

Q15. Make a first barplot of candy ranking based on winpercent values.

Answer is in code below:

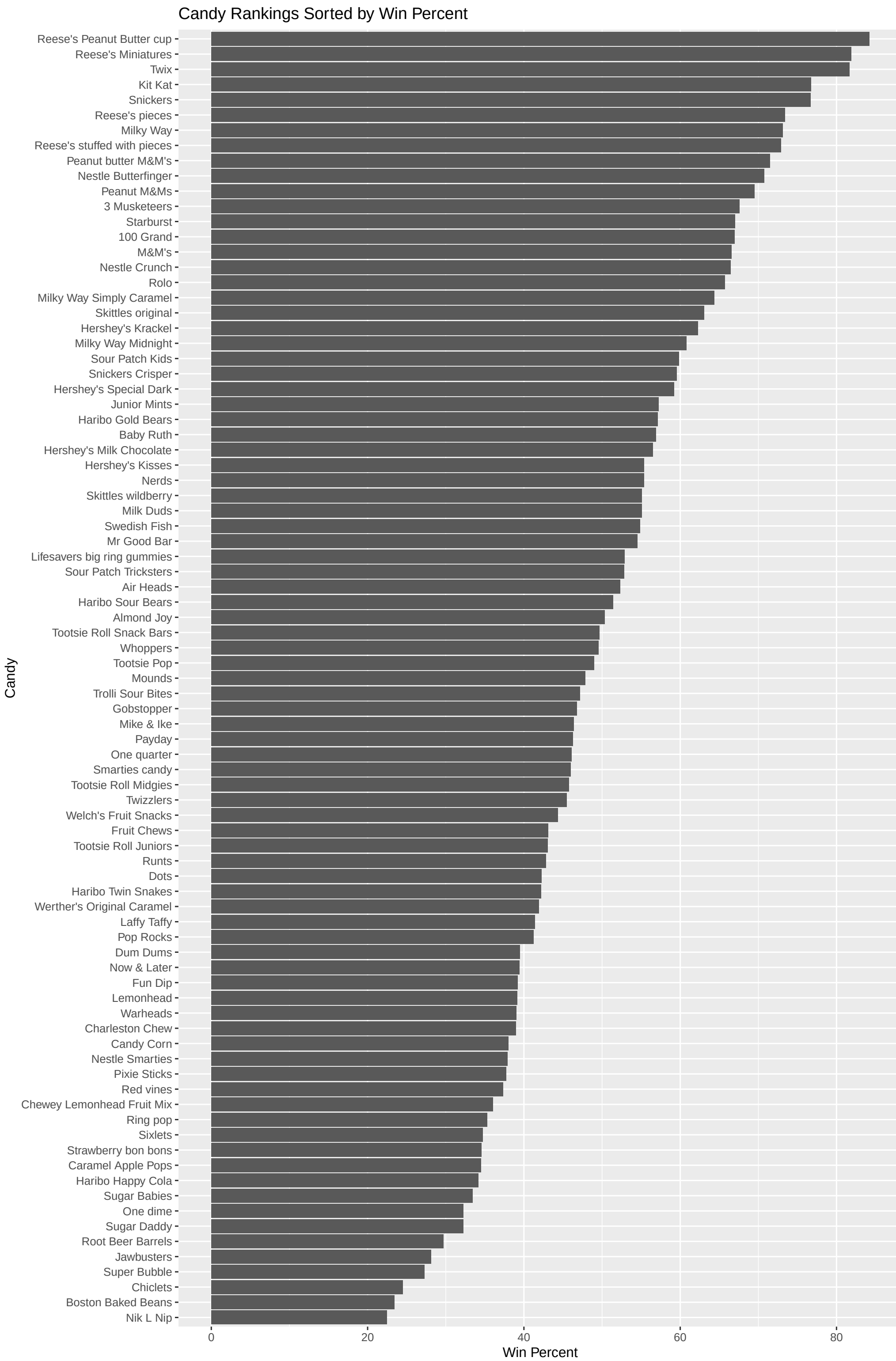
```
# Q15: Make a barplot of candy ranking based on winpercent values

# Load ggplot2
library(ggplot2)

# Make a barplot with winpercent and candy names

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```r
# Q16: Improve the barplot by sorting the bars by winpercent

# Load ggplot2
library(ggplot2)

# Make a sorted barplot of candy rankings
# reorder() sorts the candy by their winpercent

ggplot(candy) +
  aes(x = winpercent, y = reorder(rownames(candy), winpercent)) +
  geom_col() +
  labs(title = "Candy Rankings Sorted by Win Percent",
       x = "Win Percent",
       y = "Candy")
```


Candy Rankings Sorted by Win Percent
Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
Candy
0
20
40
60
80
Win Percent

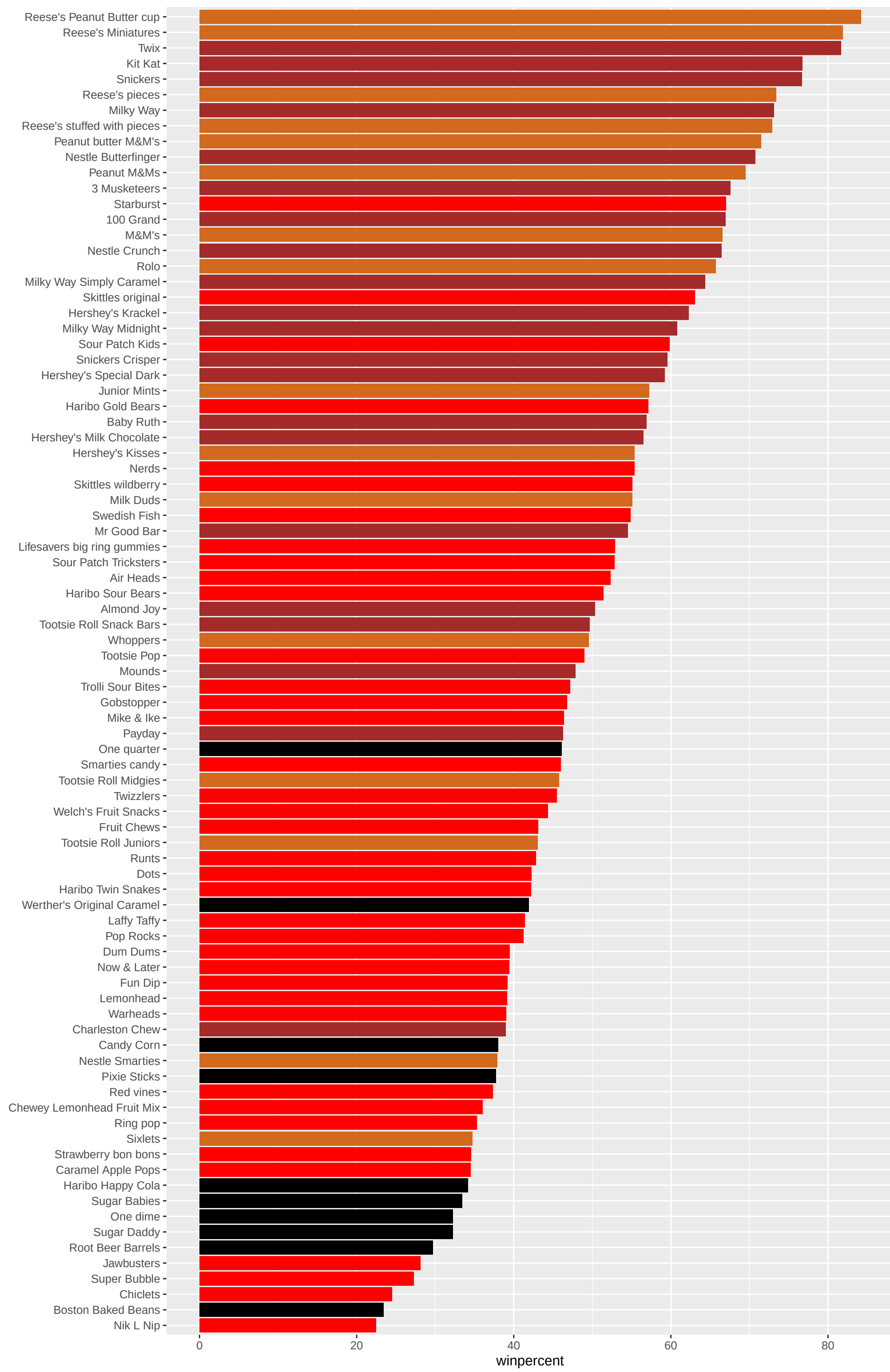
4.0.1 Time to add some useful color

Let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
# Start by making every candy black
my_cols <- rep("black", nrow(candy))

# Change candies that contain chocolate to chocolate color
my_cols[as.logical(candy$chocolate)] <- "chocolate"

# Change candies that are bars to brown
my_cols[as.logical(candy$bar)] <- "brown"

# Change candies that are fruity to red
my_cols[as.logical(candy$fruity)] <- "red"
```

Now let's try our barplot with these colors. Note that we use fill=my_cols for geom_col(). Experiment to see what happens if you use col=mycols.

```
# Make a sorted barplot of candy rankings with colors
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = my_cols) +
  ylab("")
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Now, for the first time, using this plot we can answer questions like:

> **Q17.** What is the worst ranked chocolate candy?

The worst ranked chocolate candy are Sixlets.

> **Q18.** What is the best ranked fruity candy?

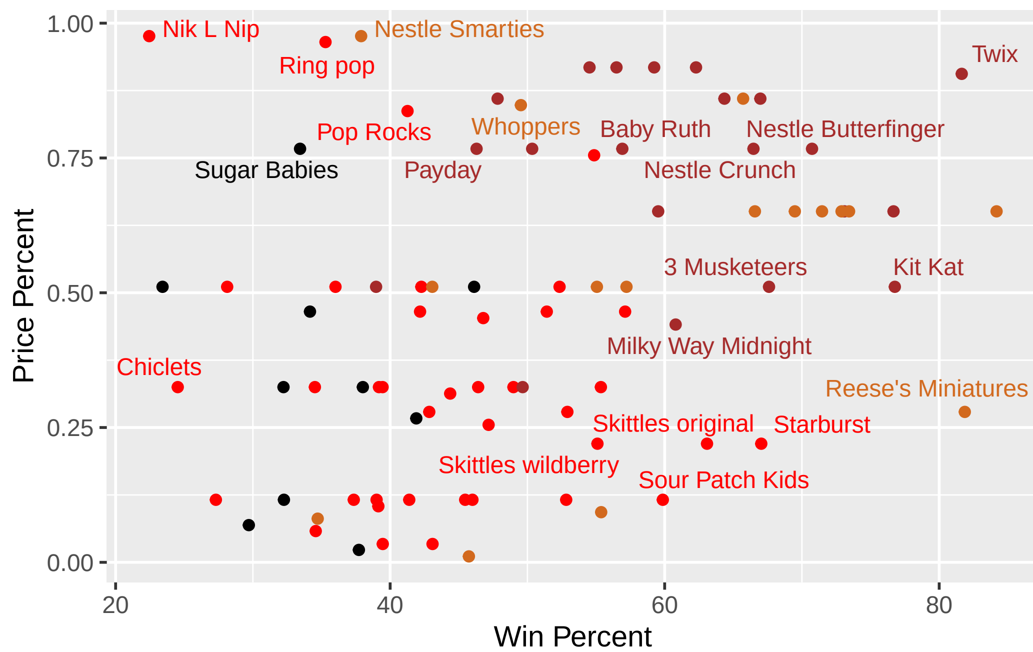
The best ranked fruity candy is starburst.

Take a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of winpercent vs the pricepercent variable. The pricepercent variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

To this plot we will add text labels so we can more easily identify a given candy. There is a regular geom_label() that comes with ggplot2. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the geom_text_repel() function from the ggrepel package.

```
#install.packages("ggrepel")

library(ggrepel)

# Make a plot
# Each point is one candy, and labels show the candy names

ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5) +
  labs(x = "Win Percent",
       y = "Price Percent")
```



> **Q19.** Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Tootsie Roll Midgets is the highest ranked in terms of winpercent for the least money. It has the highest winpercent while also having a low pricepercent.

```
# Q19: Find candies with high winpercent and low pricepercent

head(candy[order(candy$winpercent/candy$pricepercent, decreasing=TRUE), c(11,12)], n=1)
```

```
                     pricepercent winpercent
Tootsie Roll Midgies        0.011   45.73675
```

> **Q20.** What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types in the dataset are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Nik L Nip is the least popular because it has the smallest winpercent.

```
# Q20: Find the top 5 most expensive candies

ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                            pricepercent winpercent
Nik L Nip                          0.976   22.44534
Nestle Smarties                    0.976   37.88719
Ring pop                           0.965   35.29076
Hershey's Krackel                  0.918   62.28448
Hershey's Milk Chocolate           0.918   56.49050
```

Exploring the Correlation Structure

Before we dive into PCA in the next section, it's worth pausing to look more closely at the correlations between all pairs of variables.

We have already seen that being a chocolate candy tends to imply being relatively expensive and popular - in other words, these variables are positively correlated with one another. Here we will examine the magnitude and direction of all variable pair combinations to see how they relate to each other more broadly.

Side-note: This matters especially because PCA operates not on raw data but on the relationships between variables. Specifically, it finds the directions that capture the most co-variation in your data.

This is also why we will use scale=TRUE when calling prcomp() in the next section: it ensures all variables contribute equally by working from correlations (standardized and scale-free) rather than raw covariances, which would be dominated by whichever variable happens to have the largest numeric range. Can you guess which variable in our dataset that would be?

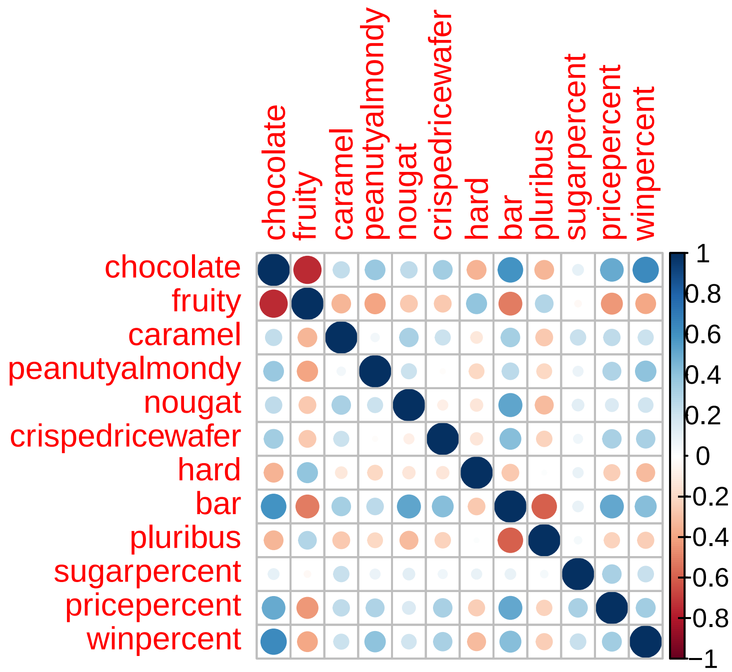
An important point here is to think of the analysis below as a preview of your upcoming PCA results: the patterns we spot here should reappear in the loadings plot at the end of the next section.

The code below calculates the pairwise correlation between all variables and views the results with the corrplot package to plot a correlation matrix.

```r
#views the results with the corrplot package to plot a correlation matrix

#install.packages("corrplot")
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



A few things to notice about this correlation matrix plot. First, the diagonal is always perfectly blue (correlation = 1.0) — every variable is perfectly correlated with itself. Second, the matrix is symmetric: the correlation between chocolate and bar is the same as between bar and chocolate, so the upper and lower triangles are mirror images. Most importantly, this matrix is exactly what PCA will decompose mathematically. When we run prcomp() on scaled data, it is performing an eigen-decomposition of this correlation matrix — finding the directions (eigen-vectors) that explain the most variance (eigen-values). The loading values you will see shortly are those eigen-vectors, and the proportion of variance explained by each PC corresponds to its eigen-value. The corrplot is therefore not just an exploratory tool, it is the structure that PCA will summarize.

> **Q22.** Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are chocolate and fruity. This means candies that are chocolate are usually not fruity, and candies that are fruity are usually not chocolate.

> **Q23.** Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Based on the corrplot, I predict that chocolate and fruity to have the largest contributions (loadings) to PC1 because they are highly negatively correlated. This means PC1 will likely separate chocolate candies from fruity candies. I would also expect bar, pricepercent, and winpercent to contribute because they appear positively associated with chocolate and help explain separation among candies.

Principal Component Analysis

Let's apply PCA using the prcomp() function to our candy dataset remembering to set the scale=TRUE argument.

Side-note: Feel free to examine what happens if you leave this argument out (i.e. use the default scale=FALSE). Then examine the summary(pca) and pca$rotation[,1] component and see that it is dominated by winpercent (which is after all measured on a very different scale than the other variables).

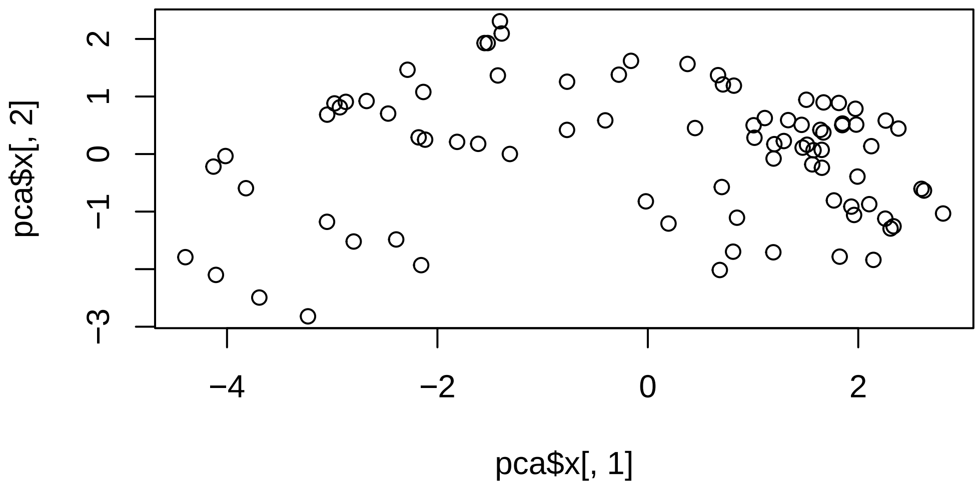
In this case we want to be sure to set `scale=TRUE` argument for `prcomp()` because we have one variable, `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

```r
#Summary of pca
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

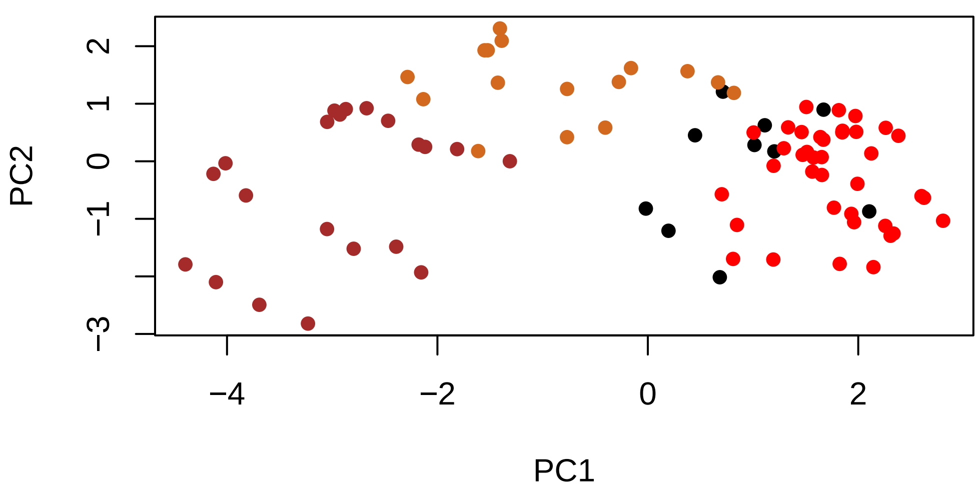
Now we can plot our main PCA score plot of PC1 vs PC2.

```r
# Plot PC1 vs PC2
plot(pca$x[, 1], pca$x[, 2])
```



We can change the plotting character and add some color:

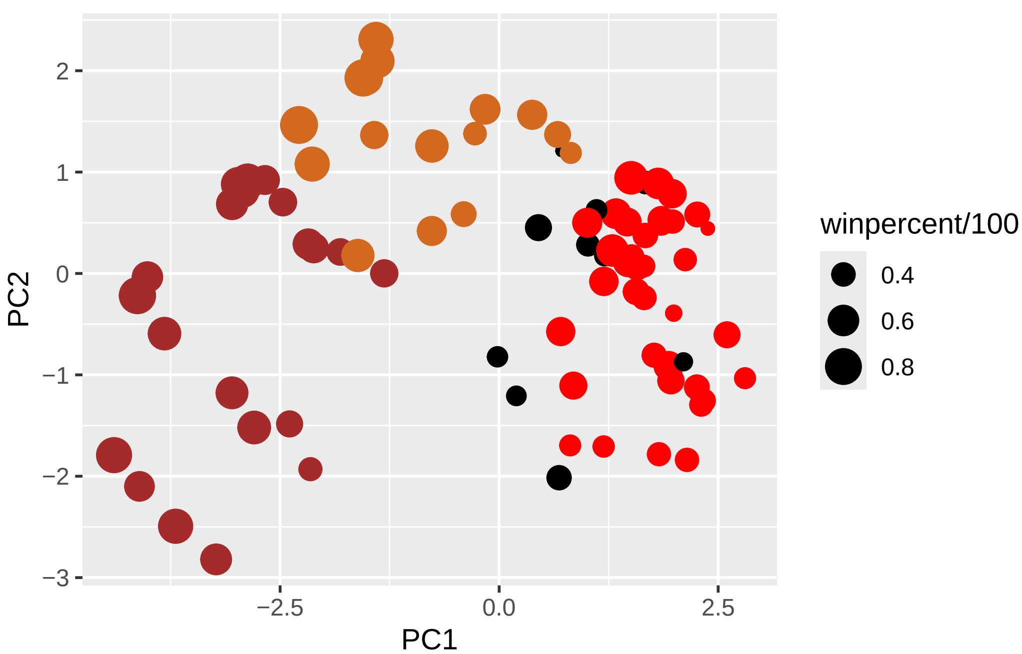
```
#Plot pca
plot(pca$x[,1:2], col=my_cols, pch=16)
```



We can make a much nicer plot with the ggplot2 package but it is important to note that ggplot works best when you supply an input data.frame that includes a separate column for each of the aesthetics you would like displayed in your final plot. To accomplish this we make a new data.frame here that contains our PCA results with all the rest of our candy data. We will then use this for making plots below

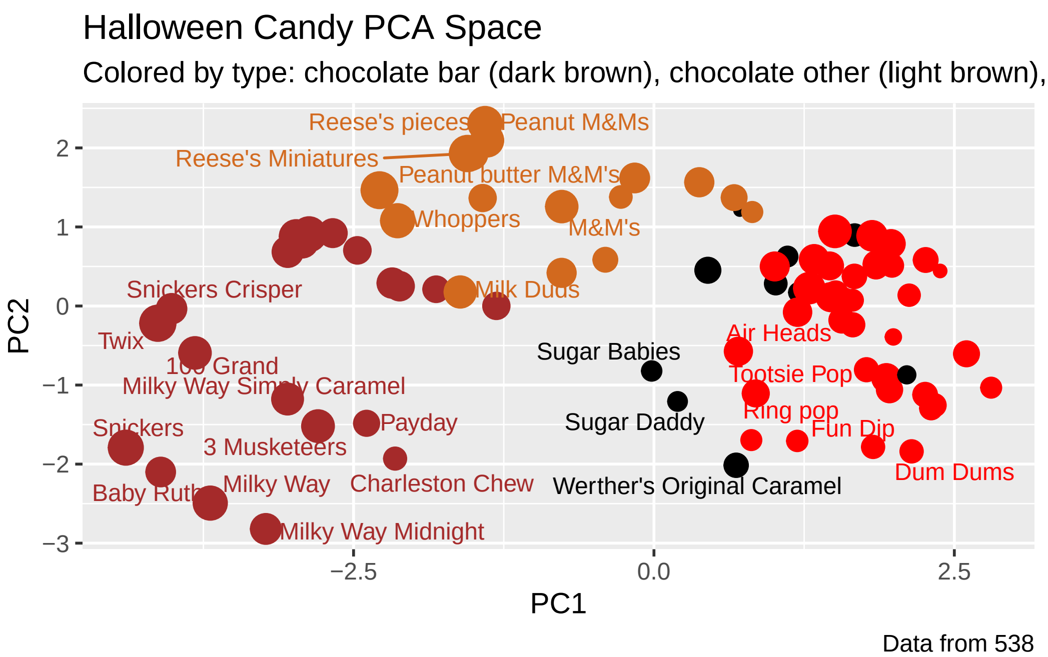
```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
my_data <- cbind(candy, pca$x[, 1:3])

# Make a ggplot PCA plot of PC1 vs PC2

p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent / 100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
p
```



Again we can use the ggrepel package and the function ggrepel::geom_text_repel() to label up the plot with non overlapping candy names like. We will also add a title and subtitle like so:

```r
library(ggrepel)

# Label the plot
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

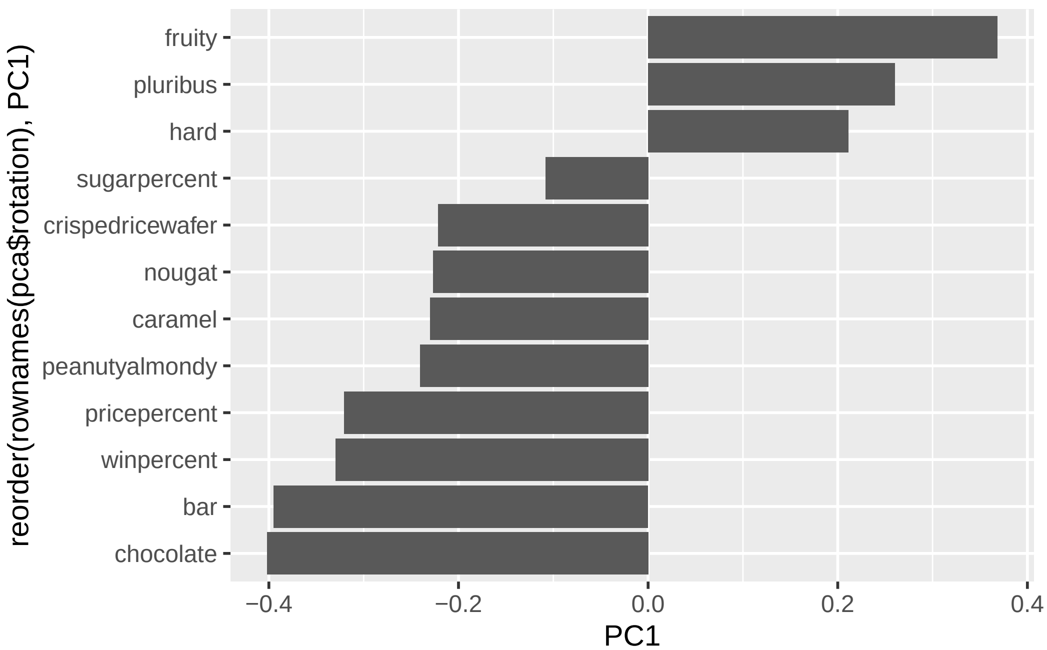


If you want to see more candy labels you can change the max.overlaps value to allow more overlapping labels or pass the ggplot object p to plotly like so to generate an interactive plot that you can mouse over to see labels:

```r
#install.packages("plotly")
#library(plotly)
#ggplotly(p)
```

Let's finish by taking a quick look at PCA our loadings. Do these make sense to you? Notice the opposite effects of chocolate and fruity and the similar effects of chocolate and bar (i.e. we already know they are correlated).

```
#PCA

ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



> **Q24.** Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. This makes sense to me because fruity candies are often different from chocolate candies, as they often come in packs or multiple pieces, which matches the positive loading for pluribus. We saw this relationship highlighted earlier in the corrplot, where chocolate and fruity were negatively correlated. Fruity in the corrplot also had strong positive association to pluribus and hard.

Summary

Starting with basic data import and exploration, you characterized the structure of the candy dataset and identified key variables. You then built increasingly sophisticated visualizations—

from simple histograms to ranked bar plots to labeled scatter plots—to reveal patterns in candy popularity and pricing.

The correlation analysis showed you how candy attributes relate to each other (chocolate and fruity are essentially mutually exclusive, for instance), and this set the stage for PCA. By reducing the original variables down to a few PCs, you created a "map of candy space" where similar candies cluster together. The PCA loadings revealed that the primary axis of variation separates chocolate-based candies from fruity ones - the same pattern we observed in the correlation matrix, now visualized in a single, interpretable plot.

> **Q25.** Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Based on my exploratory analysis, correlation findings, and PCA results, a "winning" candy appears to be chocolate-based (not fruity), often in bar form. The visualizations support this because many of the top-ranked candies had high winpercent values that were chocolate candies. The comparison between chocolate and fruity candies also showed that chocolate candies had a higher average winpercent than fruity candies. The correlation analysis supports this pattern because chocolate and fruity were negatively correlated, meaning candies are usually either chocolate or fruity, but not both. The PCA results also support this conclusion because PC1 separated chocolate/bar candies from fruity/pluribus/hard candies. Overall, these analyses complement each other by showing the same main pattern, which is that candies that are more chocolate-like tend to be more popular than candies that are more fruity-like.